

Math 63: Real Analysis

Prishita Dharampal

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Problem 1. Let S be a subset of a metric space E and let $p \in E$. Show that p is a cluster point of S if and only if p is the limit of a Cauchy sequence of points in $S \cap \{p\}^c$.

Solution.

($\implies$)

Assume p is a cluster point of $S \subset E$. Then, by definition all open ball $B_{r_i}(p)$ for some $r_i > 0$ contain an infinite number of points in S .

For each $n \in \mathbb{N}$, choose

$$p_n \in S \cap \{p\}^c \cap B_{1/n}(p)$$

Then for all n ,

$$d(p_n, p) < \frac{1}{n}$$

We claim that (p_n) is a Cauchy sequence.

That is, for a given $\epsilon > 0$, we can find an N such that $2/N < \epsilon$. Then for all $n, m \geq N$,

$$d(p_n, p_m) \leq d(p_n, p) + d(p, p_m) < \frac{1}{n} + \frac{1}{m} \leq \frac{2}{N} < \epsilon.$$

Thus (p_n) is Cauchy. Moreover, since $d(p_n, p) \rightarrow 0$, we have $\lim p_n = p$. Hence p is the limit of a Cauchy sequence of points in $S \cap \{p\}^c$.

($\impliedby$)

Assume p is the limit of a cauchy sequence of points in $S \cap \{p\}^c$. Consider an open ball $B_r(p)$, for some $r > 0$. Since, $\forall r > \epsilon > 0, \exists p_n$, such that $d(p, p_n) < \epsilon$, and there are an infinite number of $\epsilon \in (r, 0)$ there are an infinite number of points in the open ball $B_{\epsilon_N}(p)$ where $\epsilon_N = \max\{\epsilon_i, \epsilon_i \in (0, r)\}$. That is, p is a cluster point of S .

Problem 2. Let $a, b \in \mathbb{R}, a < b$. The following outlines a proof that $[a, b]$ is compact. Rewrite this proof, filling in all details:

Let $\{U_i\}_{i \in I}$ be a collection of open subsets of $\mathbb{R}$ whose union contains $[a, b]$. Let $S = \{x \in [a, b] : x > a \text{ and } [a, x) \text{ is contained in the union of a finite number of the sets } \{U_i\}_{i \in I}\}$. Then $\text{l.u.b. } S \in U_i$ for some $i \in I$. Since U_i is open we must have $\text{l.u.b. } S = b \in S$.

Solution.

Let $\{U_i\}_{i \in I}$ be a collection of open subsets of $\mathbb{R}$ whose union contains $[a, b]$. Let $S = \{x \in [a, b] : x > a \text{ and } [a, x) \text{ is contained in the union of a finite number of the sets } \{U_i\}_{i \in I}\}$. Since $[a, b] \subset \bigcup_{i \in I} U_i$, $a \in U_j$ for some $j \in I$. Since U_j is open there exists some open ball $B_r(a)$, centered at a such that $\forall x \in B_r(a)$, $d(a, x) < r$, since we are on the real line, this is just the open interval $(a - r, a + r) \in U_j$. Then, we can see that $a + r \in S$, i.e. S is not empty, and upper bounded by b , since $S \subset [a, b] \subset \bigcup_{i \in I} U_i$. Then, by the least upper bound property $\text{l.u.b. } S = y \leq b$ exists and $\text{l.u.b. } S \in U_k$ for some $k \in I$.

Because U_k is open, there exists $\epsilon > 0$ such that

$$(y - \epsilon, y + \epsilon) \subset U_k$$

Since $y = \text{l.u.b. } S$, there exists $x \in S$ with $y - \epsilon < x < y$. The interval $[a, x)$ is covered by finitely many U_i , and

$$[x, y) \subset (y - \epsilon, y) \subset U_k$$

Hence $[a, y)$ is covered by finitely many U_i , so $y \in S$. If $y < b$, then there exists $\delta > 0$ such that $y + \delta \leq b$ and

$$[y, y + \delta) \subset U_k$$

This would imply $y + \delta \in S$, contradicting that $y = \text{l.u.b. } S$. Therefore $y = b$.

Problem 3. Call a metric space sequentially compact if every sequence has a convergent subsequence. Prove that a metric space is sequentially compact if and only if every infinite subset has a cluster point.

Solution.

($\Rightarrow$)

Assume a metric space is sequentially compact. Let S be an infinite subset in this metric space. We claim that every infinite subset contains a sequence.

Proof for claim:

$S = \{s_1, s_2, s_3, \dots\}$, we can then construct a sequence (s_n) by picking some point, say s_1 from S , then picking another point from $S \setminus \{s_1\}$, say s_2 , then picking the next point from $S \setminus \{s_1, s_2\}$ and so on. Since S is infinite, this algorithm does not terminate. Hence, an infinite set contains a sequence.

Let (S_n) be a sequence in S . Then by definition of a sequentially compact metric space, (S_n) has a convergent subsequence. Let p be the point it converges to. Then p is a cluster point for the infinite set S because for any $\epsilon > 0$, we know that $\exists N, \forall n > N, d(p, p_n) < \epsilon$, or any open ball centered at p with radius some $\epsilon > 0$ contains an infinite number of points of S .

($\Leftarrow$)

Assume every infinite subset of a metric space has a cluster point. Let (S_n) be a sequence in E . Consider the set $S = \{s \in S_n\}$, here we have two cases:

1. (S_n) has an infinite number of distinct points and S is infinite. Since every infinite set has a cluster point, $p \in S$. Pick n_1 such that p_{n_1} is in the open ball $B_1(p)$ of center p and radius 1, then pick an integer $n_2 > n_1$ such that p_{n_2} is in the open ball $B_{1/2}(p)$, then pick an integer $n_3 > n_2$ such that p_{n_3} is in the open ball $B_{1/3}(p)$; this process can be continued indefinitely since each open ball of center p contains an infinite number of points of S . We end up with a subsequence of $p_1, p_2, p_3, \dots$ whose n^{th} term has distance less than $1/n$ from p , for all $n \geq 1$. This subsequence converges to p .
2. (S_n) has a finite number of distinct points. Then at least one point $p \in S$ occurs an infinite number of times in (S_n) . Then the subsequence $p, p, p, \dots$ is a subsequence that converges to p .

Either ways, every sequence has a convergent subsequence.

Therefore, a metric space is sequentially compact if and only if every infinite subset has a cluster point.

Problem 4. Call a metric space totally bounded if, for every $\epsilon > 0$, the metric space is the union of a finite number of closed balls of radius ϵ . Prove that a metric space is totally bounded if and only if every sequence has a Cauchy subsequence.

Solution.

($\implies$)

Assume a metric space E is totally bounded. That is E is the union of a finite number of closed ϵ —balls. If a sequence $(p_n) \in E$ exists then one of these balls, say B_1 , has an infinite number of points. Again, because E is totally bounded, B_1 is the union of a finite number of closed $\epsilon/2$ —balls, and one of these balls B_2 has an infinite number of points. We can keep repeat this process, and we get,

$$E \supset B_1 \supset B_2 \supset B_3 \cdots$$

Pick a point p_i from $(p_n) \cap B_i$, for all $i \geq 1$. This forms a Cauchy subsequence, since for all $j, k > i$ we have $d(p_j, p_k) < \epsilon/i$. Since we started with an arbitrary sequence, this works for any sequence in the metric space, hence if a metric space is totally bounded then every sequence has a Cauchy subsequence.

($\impliedby$)

We will prove the statement by proving it's contrapositive. Assume E is not totally bounded. Then there exists some ϵ such that E is the union of an infinite number of closed ϵ —balls.

$$E = \bigcup_{i=1}^{\infty} B_i$$

Now we construct a sequence by picking the centers of all the closed infinite ϵ —balls. Since each ϵ —ball only contributes one point to the sequence this sequence doesn't have a Cauchy subsequence. Hence proving the contrapositive.

Problem 5. Prove that the following three conditions on a metric space E (cf. Probs. 35, 36) are equivalent:

1. E is compact.
2. E is sequentially compact.
3. E is totally bounded and complete.

(Hint: That (1) implies (2) occurs in the text. That (2) implies (3) is easy. That (3) implies (1) follows from the argument of the last proof of § 5.)

Solution.

1. (1) implies (2).

Assume E is a compact metric space. And any sequence of points in a compact metric space has a convergent subsequence (by Corollary 1 of the Theorem on page 56), that is E is sequentially compact.

2. (2) implies (3).

Assume E is a sequentially compact metric space. Then, by definition, every sequence in E has a convergent subsequence. Convergent sequences in a metric space are Cauchy (Proposition, p. 51). Then, by Problem 4, we know that E is totally bounded. We also know that the Cauchy sequences converge to points in E , hence E is complete. Thus, if a metric space is sequentially compact it is totally bounded and complete.

3. (3) implies (1).

Assume E is totally bounded and complete.

We prove that E is compact by contradiction. Assume E is not compact, then there is a collection of open subsets $\{\mathcal{U}\} = \{\mathcal{U}_i\}_{i \in I}$ of E whose union contains E , but such that the union of no finite subcollection of $\{\mathcal{U}\}$ contains E . Since E is totally bounded, it is in the union of a finite number of closed balls $B_1, B_2, \dots, B_r$ of fixed radius, say $1/2$. Then,

$$E = (E \cap B_1) \cup (E \cap B_2) \cup \dots \cup (E \cap B_r)$$

At least one of the sets $E \cap B_1, \dots, E \cap B_r$, say $E \cap B_j$ is not contained in the union of any finite subcollection of $\{\mathcal{U}\}$. $E_1 = E \cap B_j$ is closed and bounded and $\forall p, q \in E_1$, $d(p, q) \leq 1$. Again since E is totally bounded, we know that E_1 is the union of a finite number of closed balls of radius $1/4$, repeated the above argument to get a closed subset E_2 of E_1 that is not contained in the finite subcollection of $\{\mathcal{U}\}$. Repeating the argument with $1/6, 1/8, 1/10, \dots$ we get the following sequence of closed sets,

$$E \supset E_1 \supset E_2 \supset E_3 \dots$$

none of which are contained in a finite subcollection of $\{\mathcal{U}\}$, and such that $\forall p, q \in S_N, d(p, q) \leq 1/N$, for each $N = 1, 2, 3, \dots$. Let $p_N \in S_N$.

The sequence

$$p_1, p_2, p_3, \dots$$

is a Cauchy since if for $x, y > N$, such that $p_x, p_y \in S_N$, then $d(p_x, p_y) < 1/N$. Since E is complete, the sequence converges to some point p in E . Hence, $p \in \mathcal{U}_i \in \{\mathcal{U}\}$. Since $\mathcal{U}_i$ is open there is some ball $B_\epsilon(p)$ of radius ϵ centered at p in it. Pick some N such that $1/N < \epsilon/2$ and $d(p, p_N) < \epsilon/2$. Then for any point $p_0 \in E_N$ we have

$$d(p, p_0) \leq d(p, p_N) + d(p_N, p_0) < \epsilon/2 + 1/N < \epsilon$$

Thus, E_N is contained in the open ball of center p and radius ϵ , i.e. $E_N \subset \mathcal{U}_i$. This contradicts the statement that E_N is not contained in the union of any finite subcollection of $\{\mathcal{U}\}$.

Thus, E is compact.